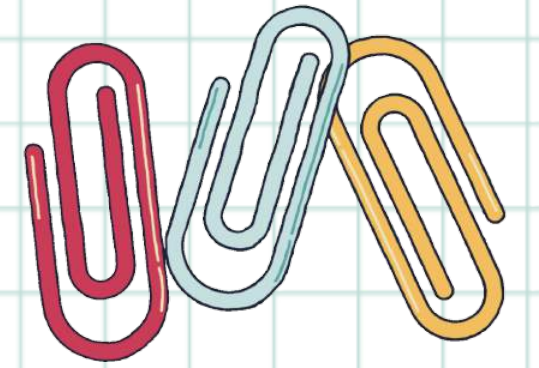
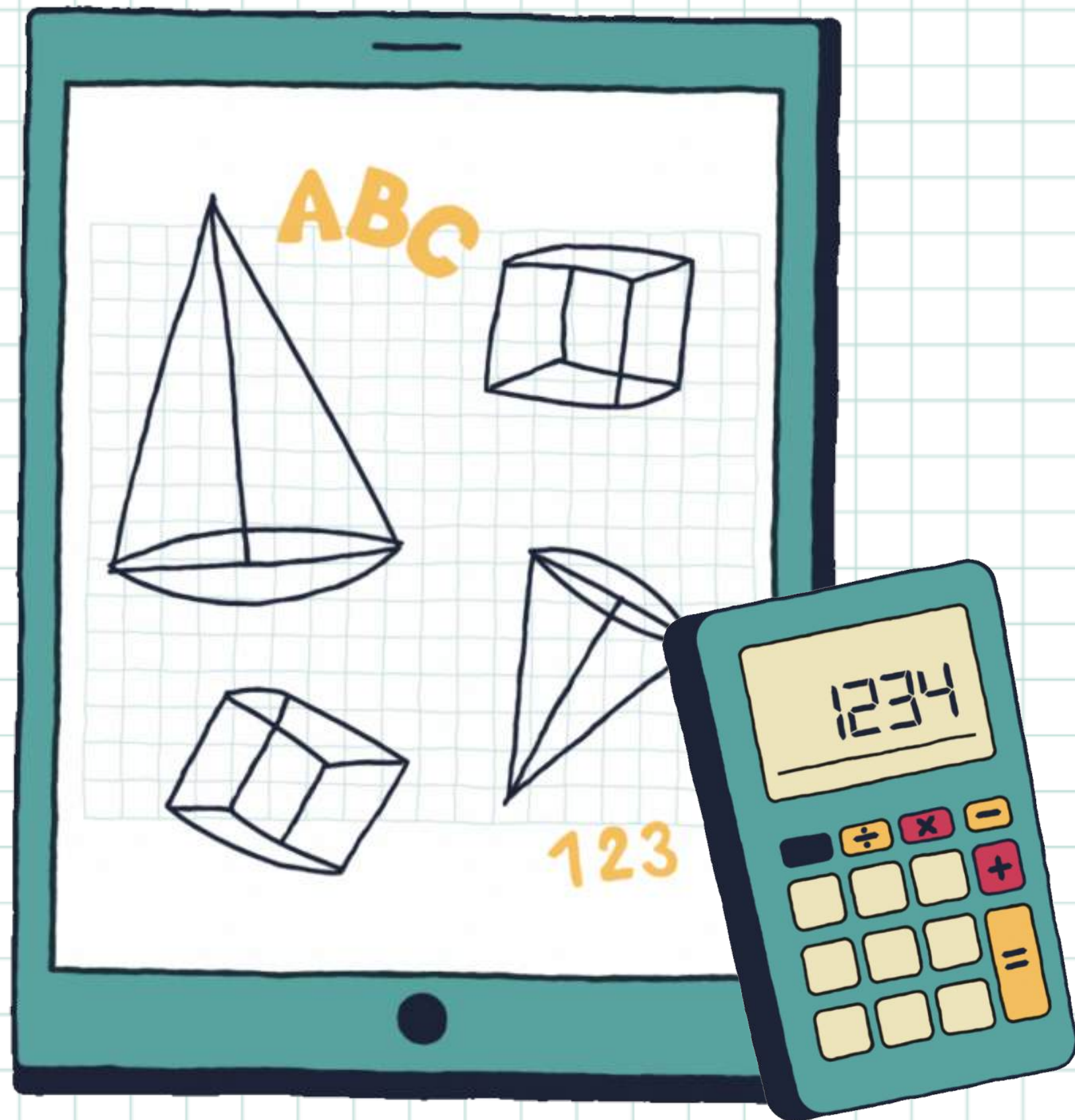
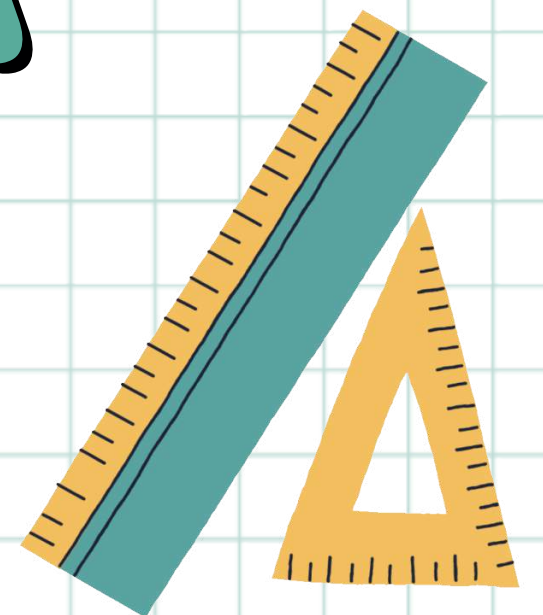
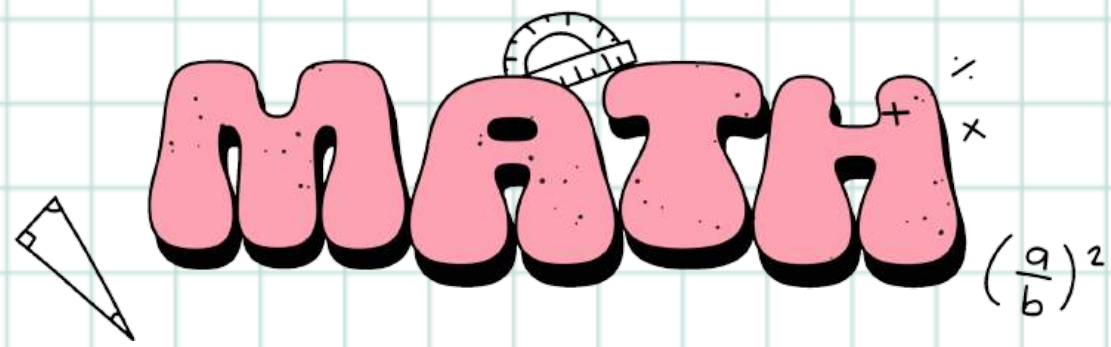




AREA – COUNTING SQUARES & FORMULA

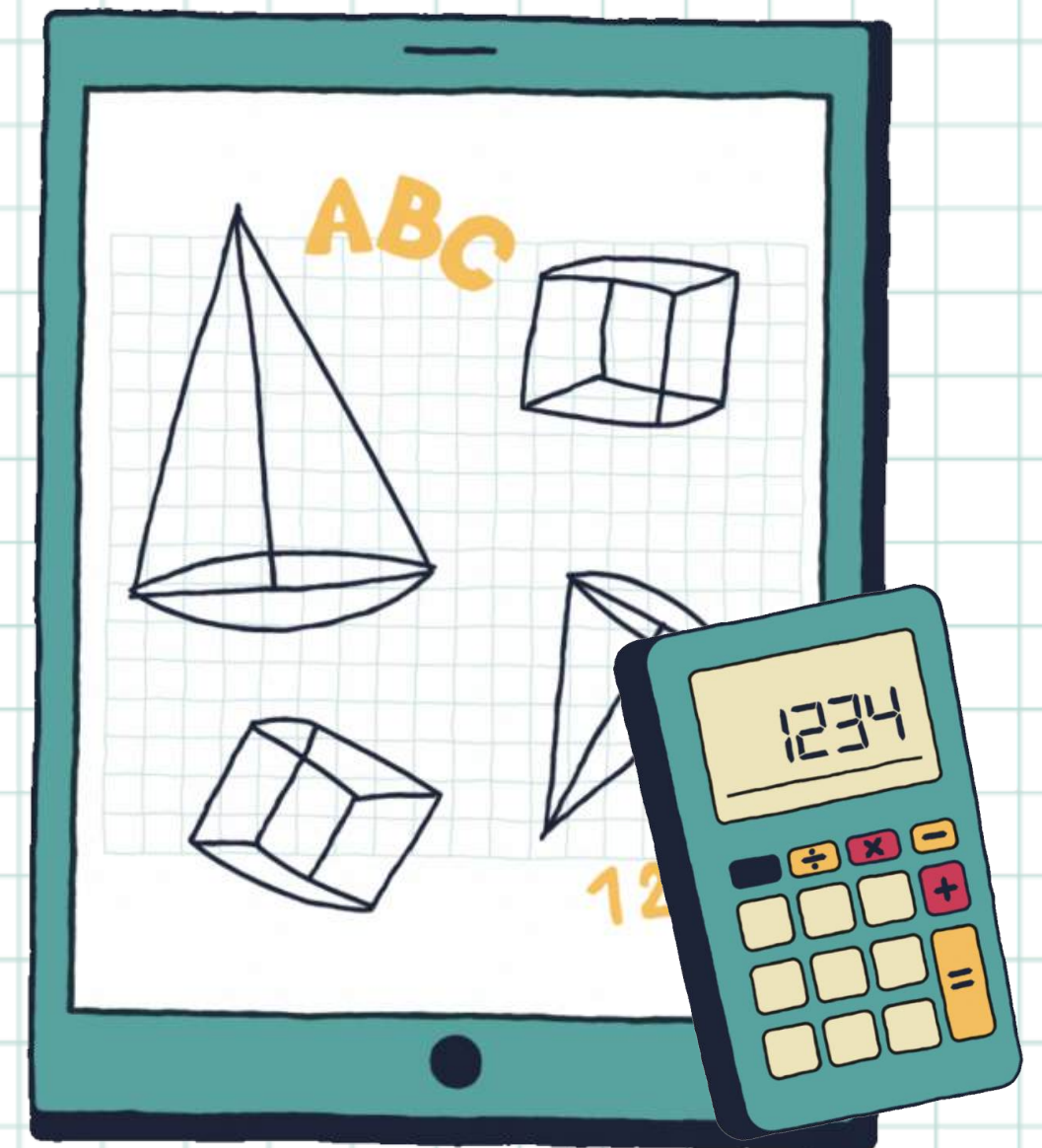


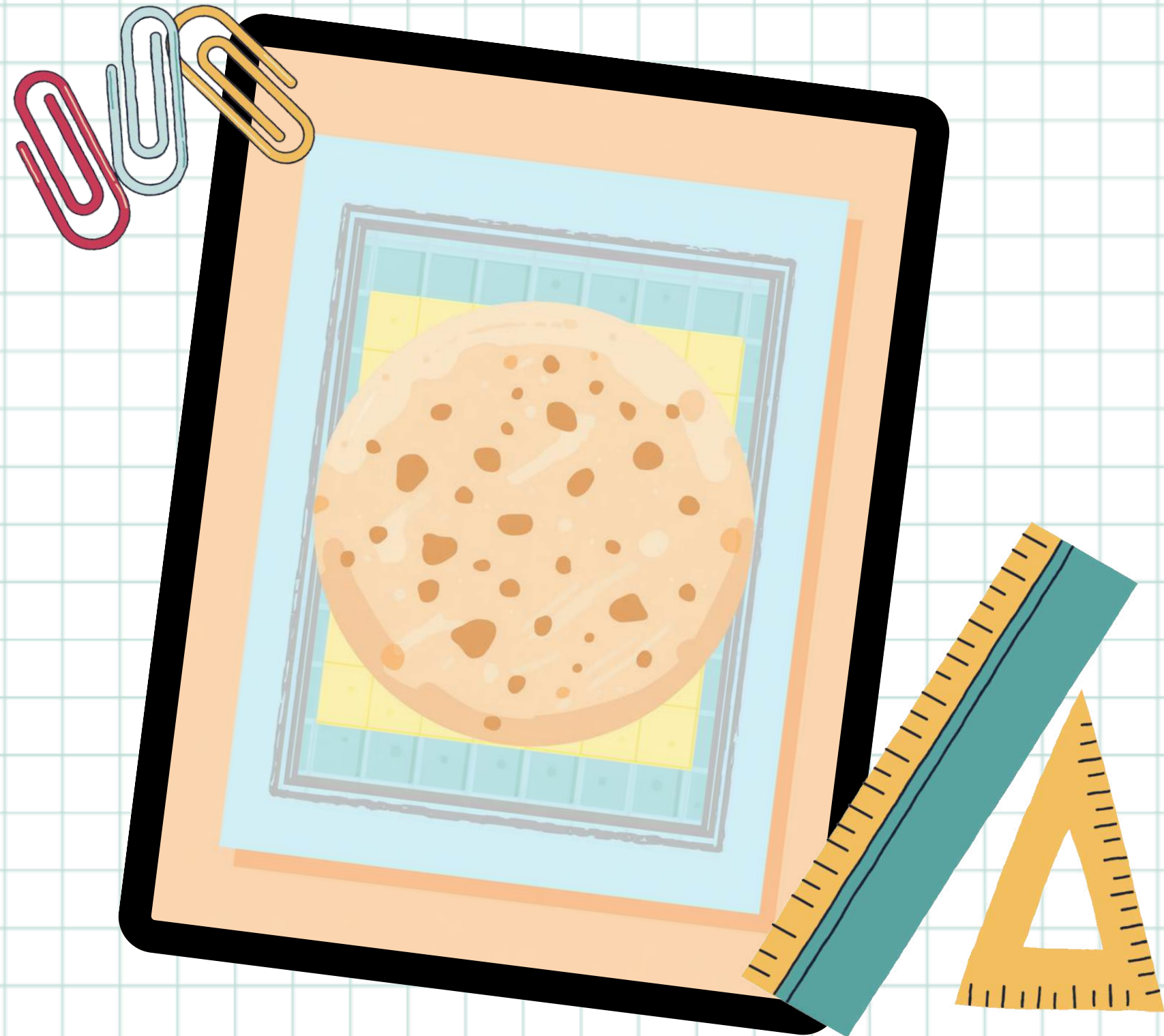
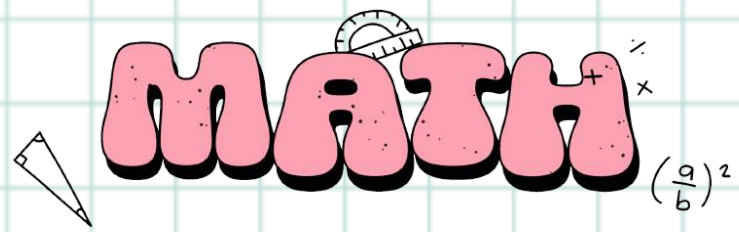


The Chapati Challenge

Dadi is making chapatis in her kitchen. Look at the tava — there are TWO chapatis! One is BIG and one is SMALL. Think and answer: Which chapati covers MORE space on the plate?

That's AREA! The bigger chapati takes up more space — it has a bigger area.
Today we will learn how to MEASURE area. Let's begin!



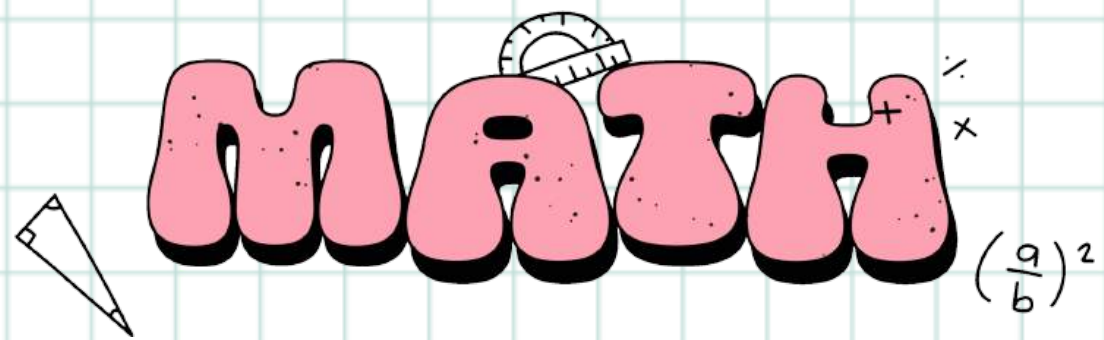


What is Area?

Area is how much flat space a shape **COVERS**!
Think of the Chapati Test: A bigger chapati covers more space on your plate — that means it has a bigger area!

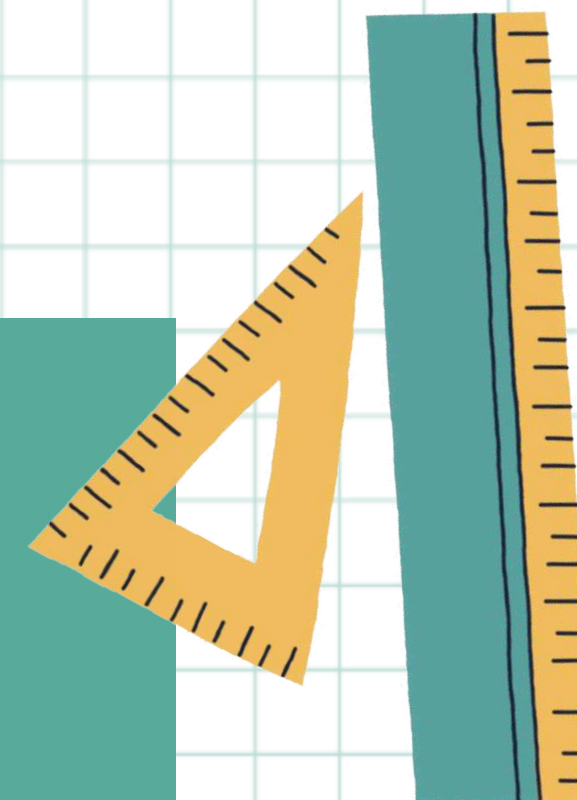
When we press a chapati onto grid paper, we can see how many squares it covers. That's how we measure area!

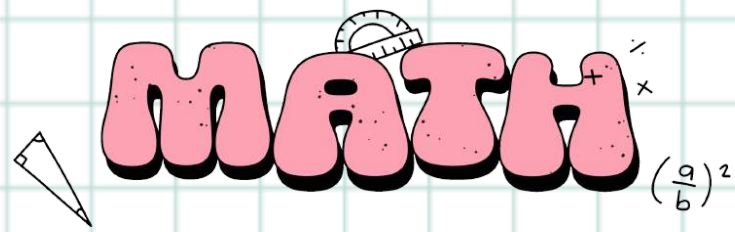
Key Point: Area = The Space **INSIDE** a Shape



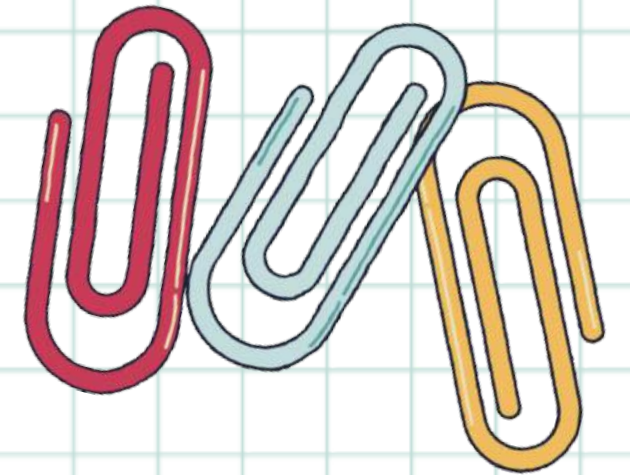
Unit Squares

A Unit Square is a square that is 1 unit long and 1 unit wide.
Area of one unit square = 1 square unit (1 sq cm or 1 cm²)
Think of it like one small tile on the kitchen floor — that's exactly one unit square!

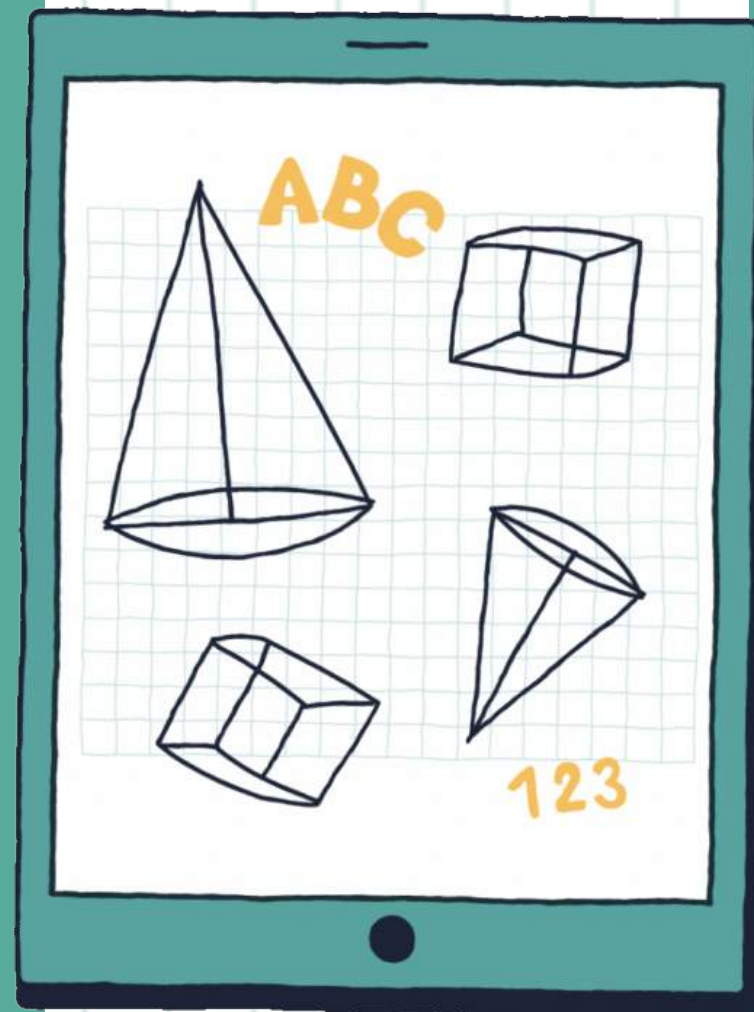




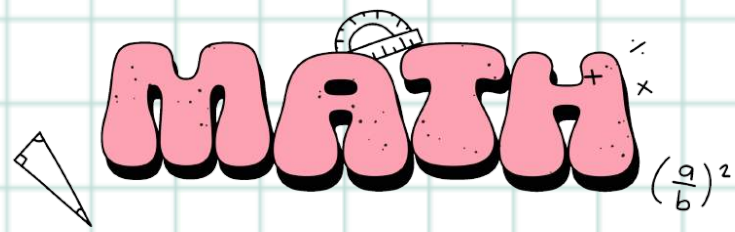
Counting Squares Method



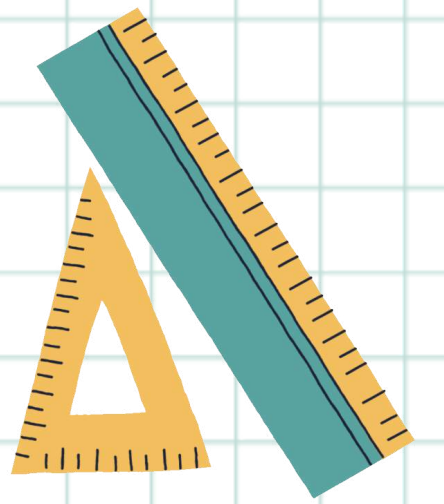
How to Find Area by Counting:
Look at the rectangle on the grid.
Count each small square inside!
1, 2, 3, 4...
5, 6, 7, 8...
9, 10, 11, 12!
We counted 12 unit squares inside
the rectangle.



Area = 12 sq cm
This rectangle is 3 units tall and 4
units wide.
3 rows $\times$ 4 columns = 12 squares
✓ Remember: Always write the
UNIT!
Use sq cm or cm^2 after your
number.
Without the unit, your answer is
incomplete!



Practice Counting

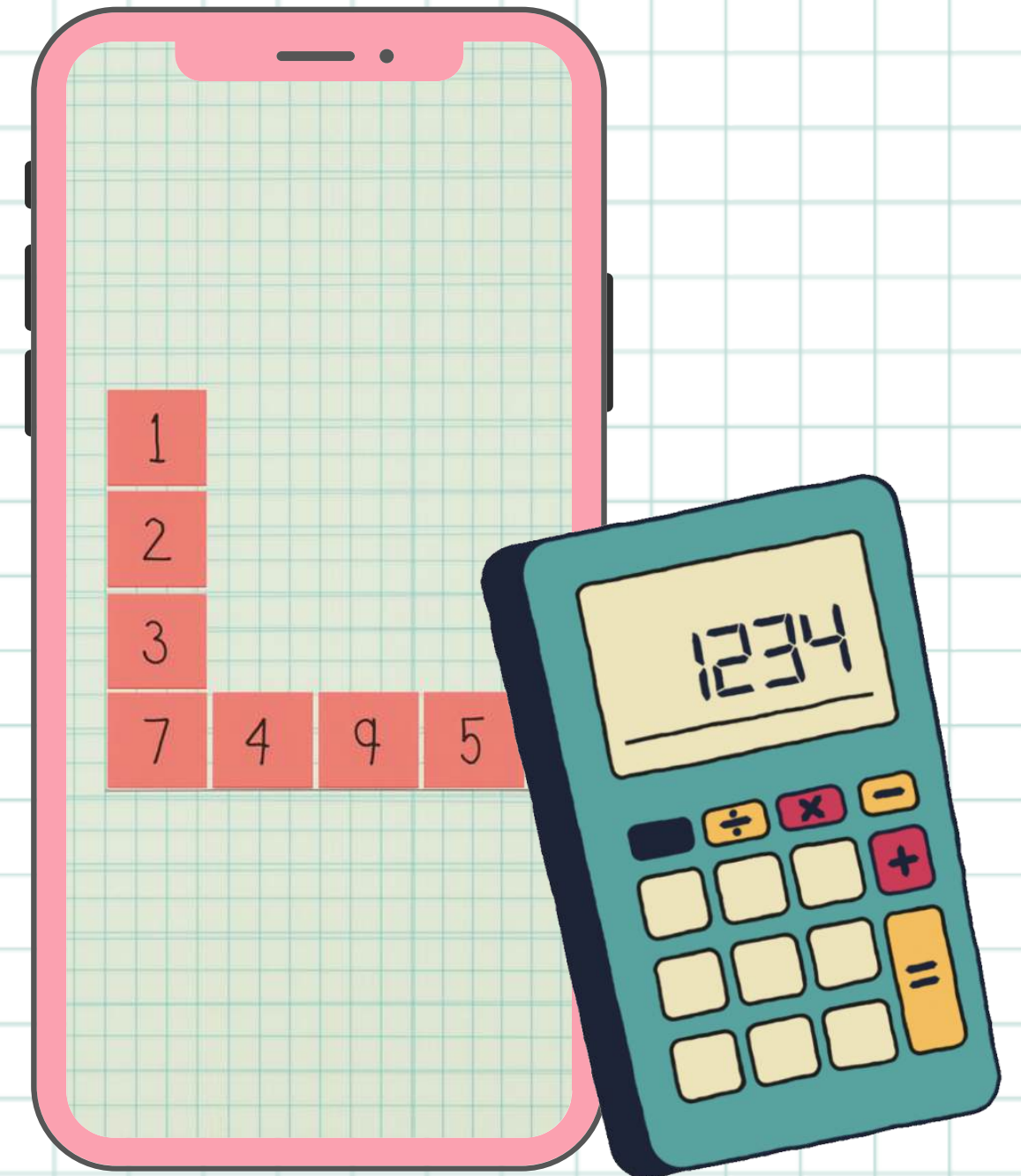


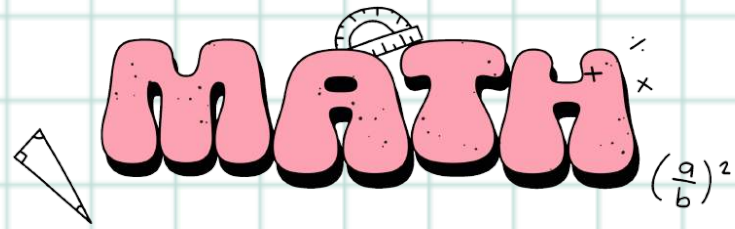
ONE

Look at the L-shaped figure on the grid. Let's count the unit squares together, row by row! Count with me: 1, 2, 3, 4, 5, 6, 7, 8 unit squares!

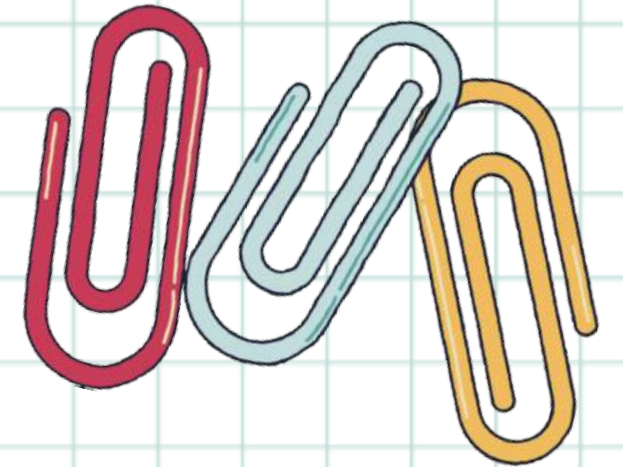
TWO

Area = 8 sq cm! Remember to always write the UNIT after your number. Great job counting all the squares inside the shape!





Same Area, Different Shapes

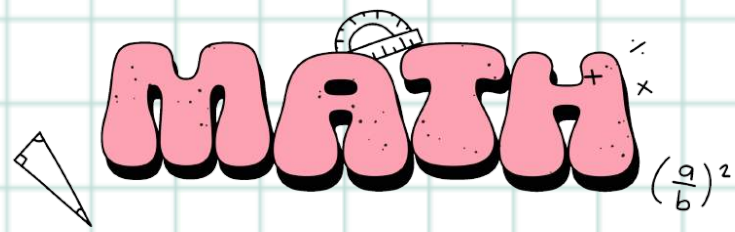


SHAPE A: RECTANGLE

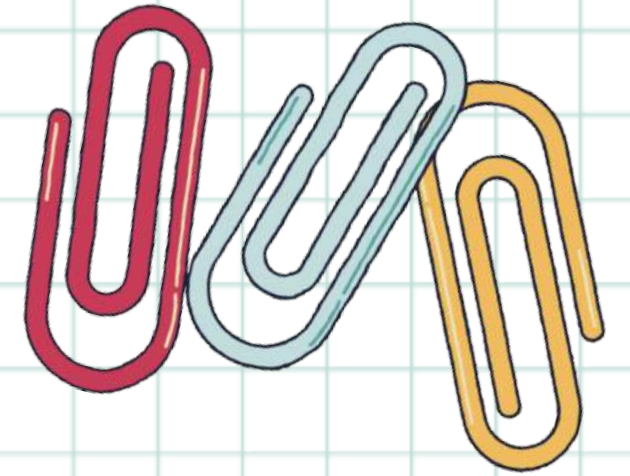
This rectangle has 10 unit squares inside. Count them: 2 rows $\times$ 5 columns = 10 squares. Even though it looks long and flat, the area is 10 sq cm!

SHAPE B: L-SHAPE

This L-shape also has 10 unit squares inside! It looks completely different from the rectangle, but the area is the same: 10 sq cm. Area is about INSIDE SPACE!



The Formula — Rows × Columns

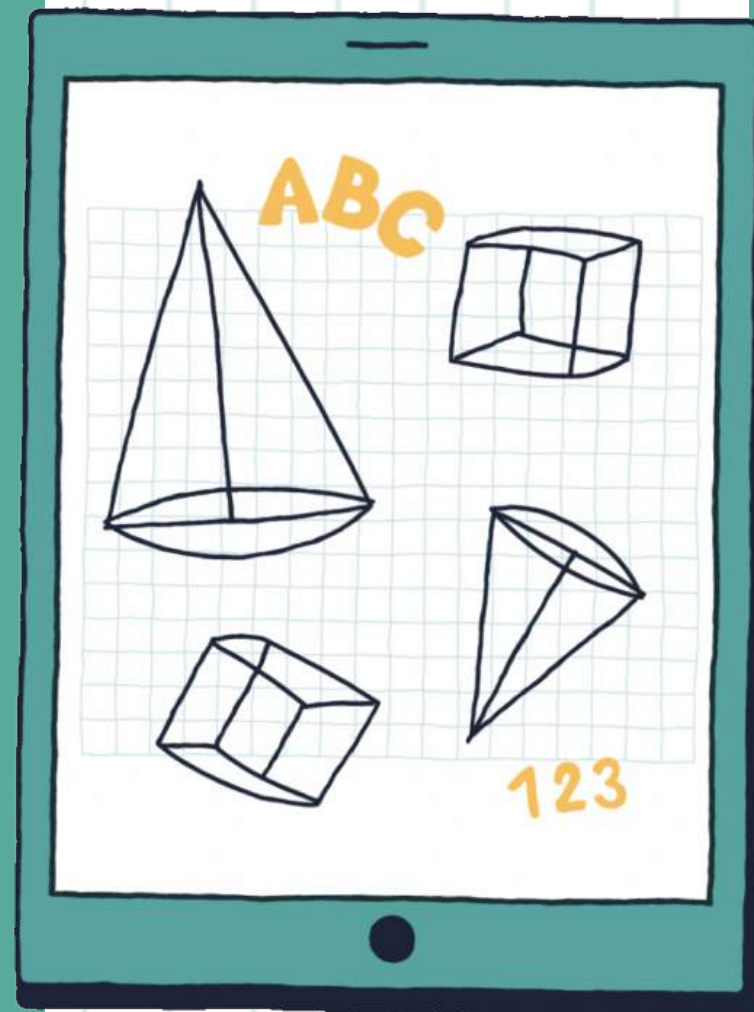


Look at this rectangle! It has 3 rows and 4 columns.
Let's count each row:

- Row 1 = 4 squares
- Row 2 = 4 squares
- Row 3 = 4 squares

We can add them:

$$4 + 4 + 4 = 12 \text{ squares!}$$



But wait — there's a **SHORTCUT!**
Instead of adding, we can **MULTIPLY:**

$$3 \text{ rows} \times 4 \text{ columns} = 12$$

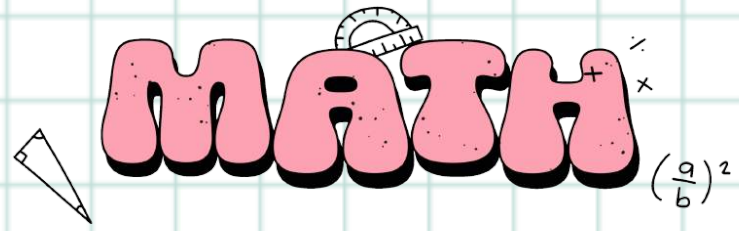
So the formula is:

$$\text{Area} = \text{Length} \times \text{Width}$$

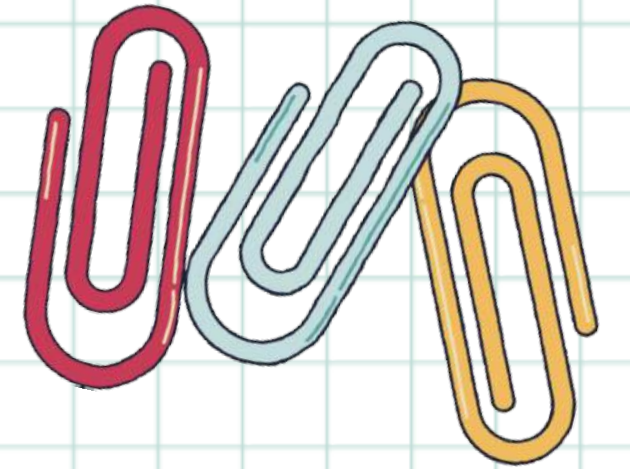
For our rectangle:

$$\text{Area} = 4 \times 3 = 12 \text{ sq cm}$$

Much faster, right? 🎉



Area Formulas



RECTANGLE FORMULA

Area = Length $\times$ Width

Example: Classroom floor

$6\text{m} \times 4\text{m} = 24 \text{ sq m}$

24 square tiles (each $1\text{m} \times 1\text{m}$) cover the whole floor!

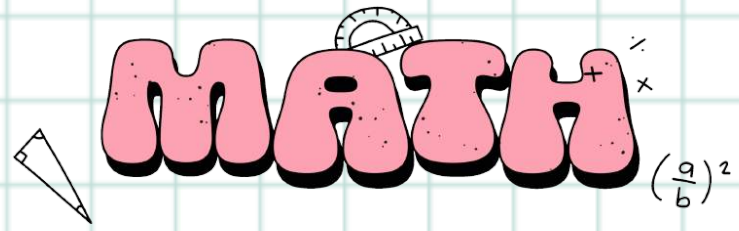
SQUARE FORMULA

Area = Side $\times$ Side

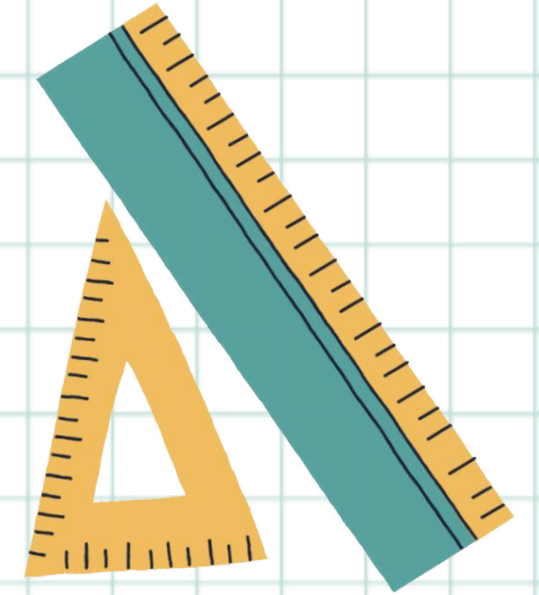
Example: Window pane

$2\text{m} \times 2\text{m} = 4 \text{ sq m}$

Remember: All sides of a square are equal!



Area vs Perimeter



Let's understand the difference!

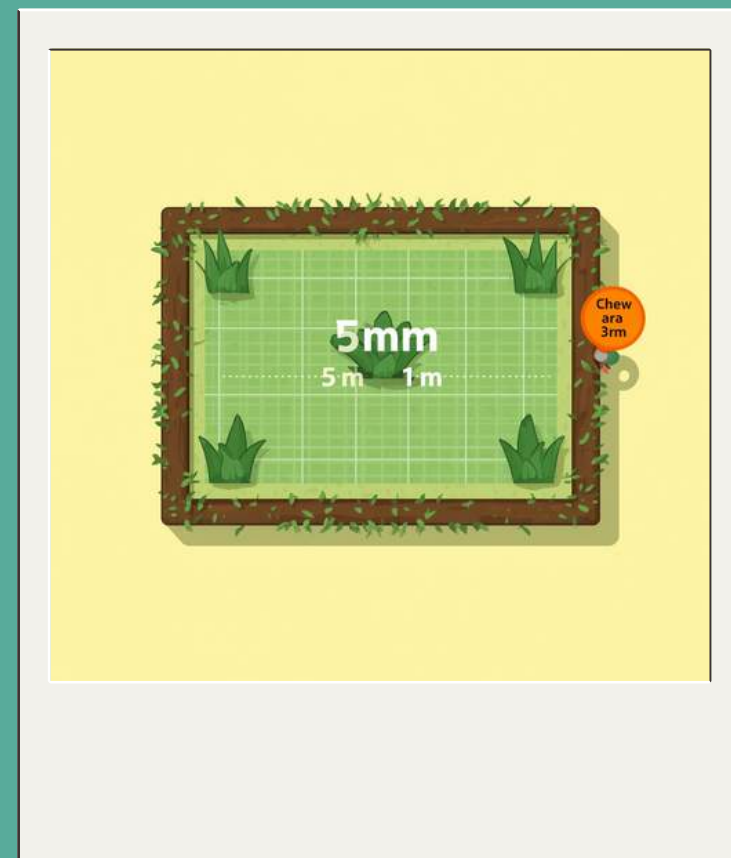
🌿 **AREA** = grass **INSIDE** garden
(The space inside the shape)

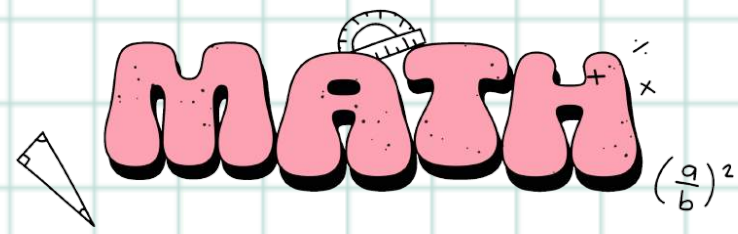
Area = $5 \times 3 = 15$ sq m

🏗️ **PERIMETER** = fence **AROUND** garden
(The distance around the shape)

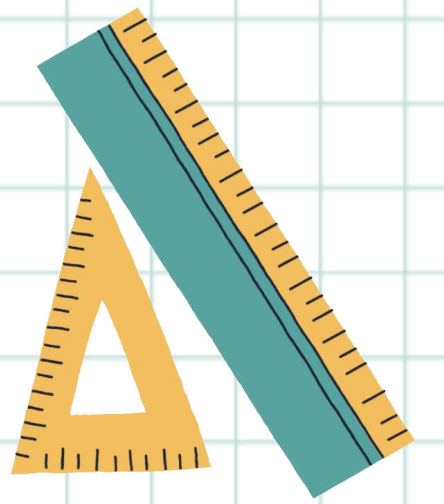
Perimeter = $5 + 3 + 5 + 3 = 16$ m

⚠️ Remember: Different values, different units — never mix them up!





Real-Life Indian Examples



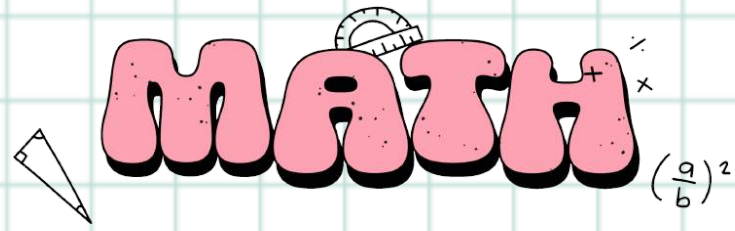
HOME

- 🏠 Mummy buying floor tiles — area of kitchen
- 📖 Priya decorating notebook cover — area for sticker paper
- 🌱 Mali planting grass — area of school ground

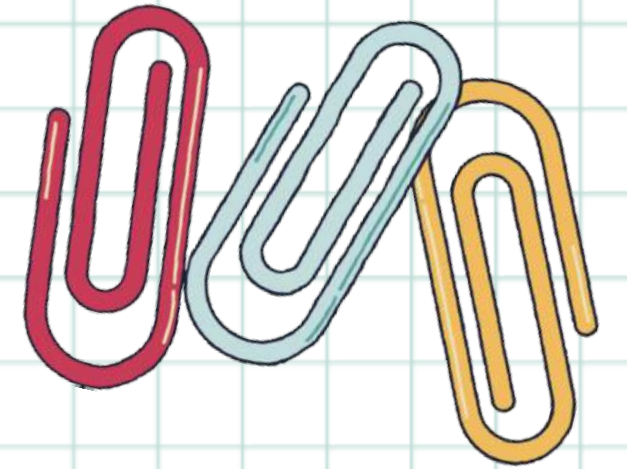
FUN!

- 🪔 Rangoli design for Diwali — area of floor covered
- 🏏 Cricket pitch turf measurement — area needed
- 💡 Area tells us how much space something covers!





Key Vocabulary



AREA & MEASUREMENT

AREA

Space inside a shape

UNIT SQUARE

1 cm $\times$ 1 cm square

SQUARE UNITS

sq cm, cm², sq m, m²

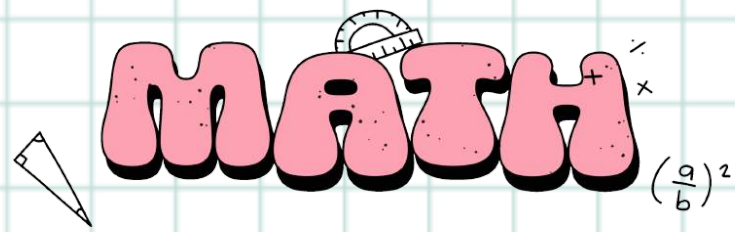
LENGTH & WIDTH

LENGTH

How long a shape is

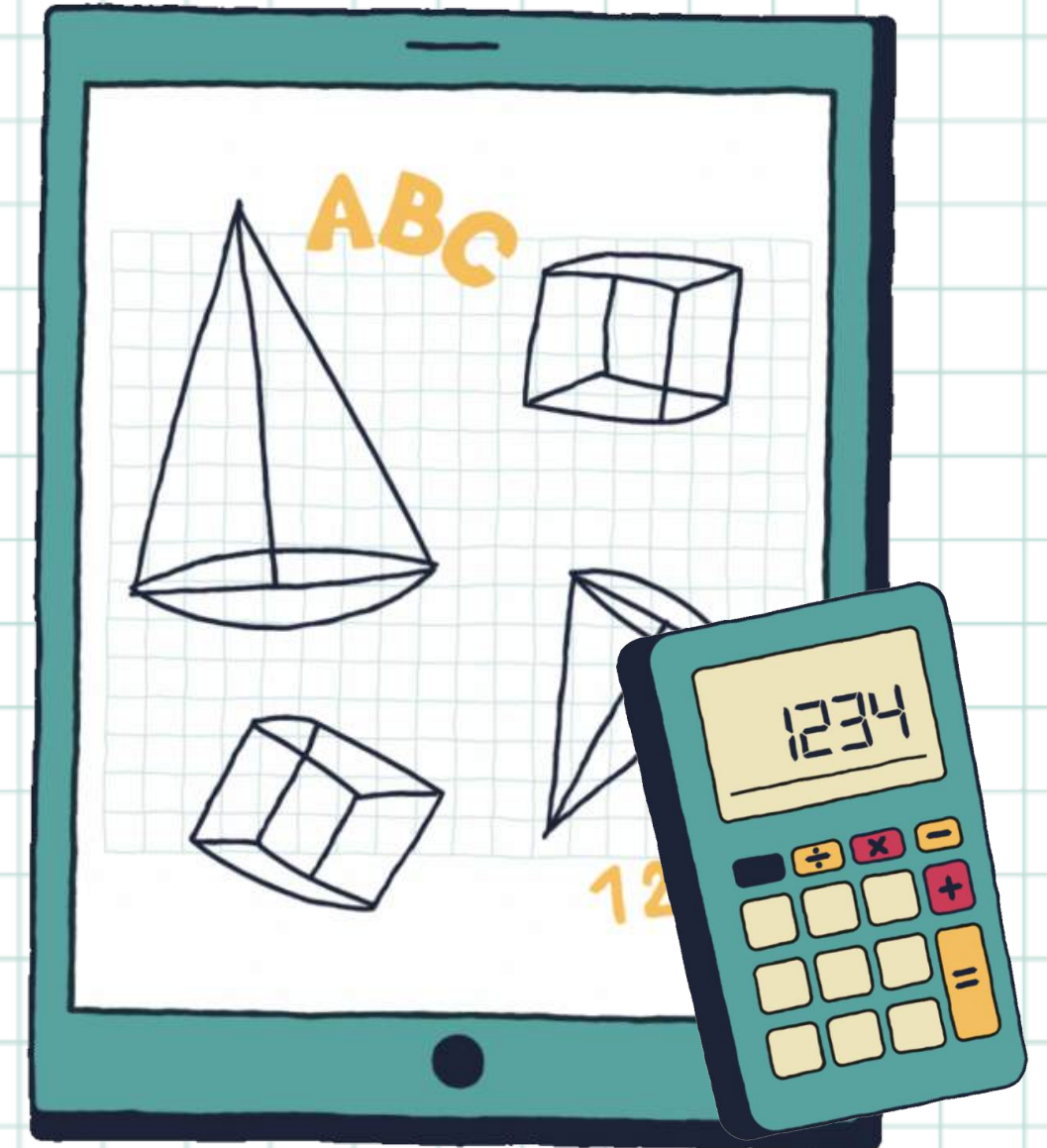
WIDTH

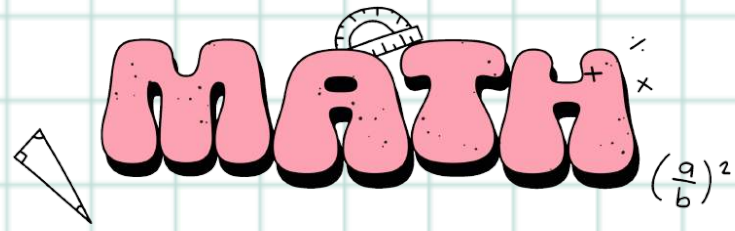
How wide a shape is



REMEMBER! 3 Ways to Find Area

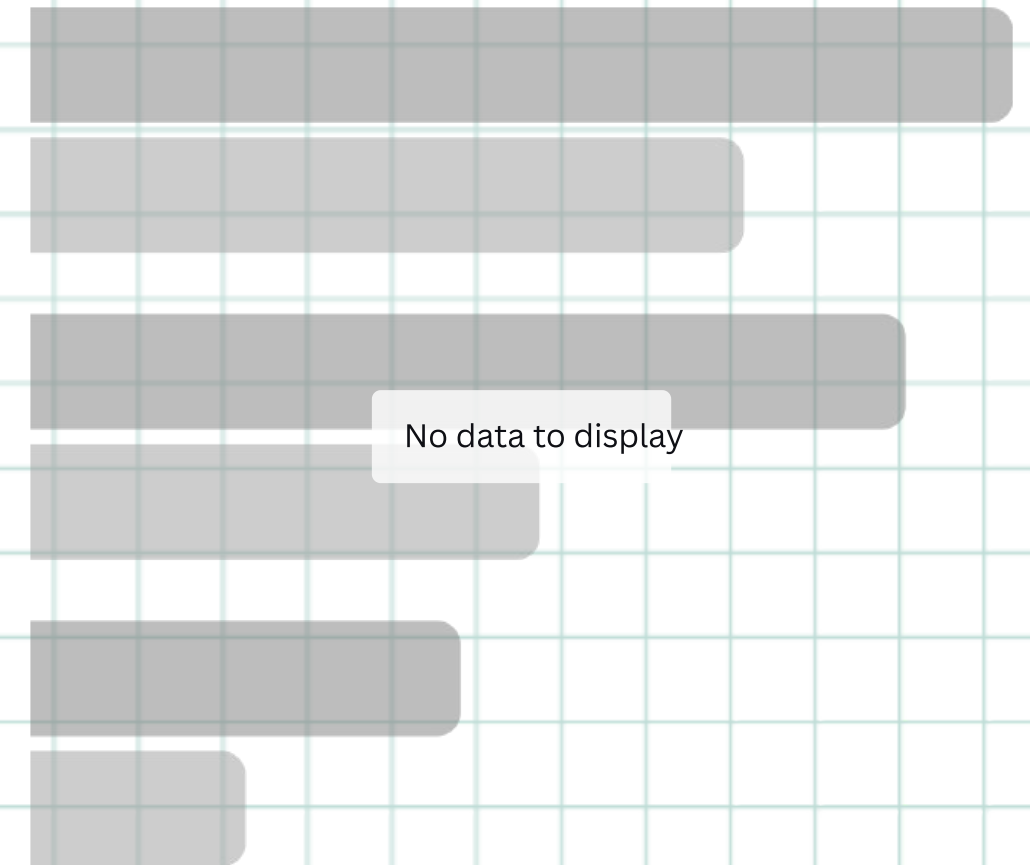
- ✓ Count the unit squares inside the shape
Look at the grid and count each small square that fits inside your shape. Each square = 1 square unit!
- ✓ Use the formula: Length $\times$ Width
For rectangles and squares, multiply the length by the width. It's a shortcut! Example: $4 \text{ cm} \times 3 \text{ cm} = 12 \text{ sq cm}$
- ✓ Always write the UNIT (sq cm or m^2)
Never forget to add "sq cm" or " cm^2 " after your answer. Without units, your answer is incomplete!
Shababsh! You've got this! Area Expert 🌟



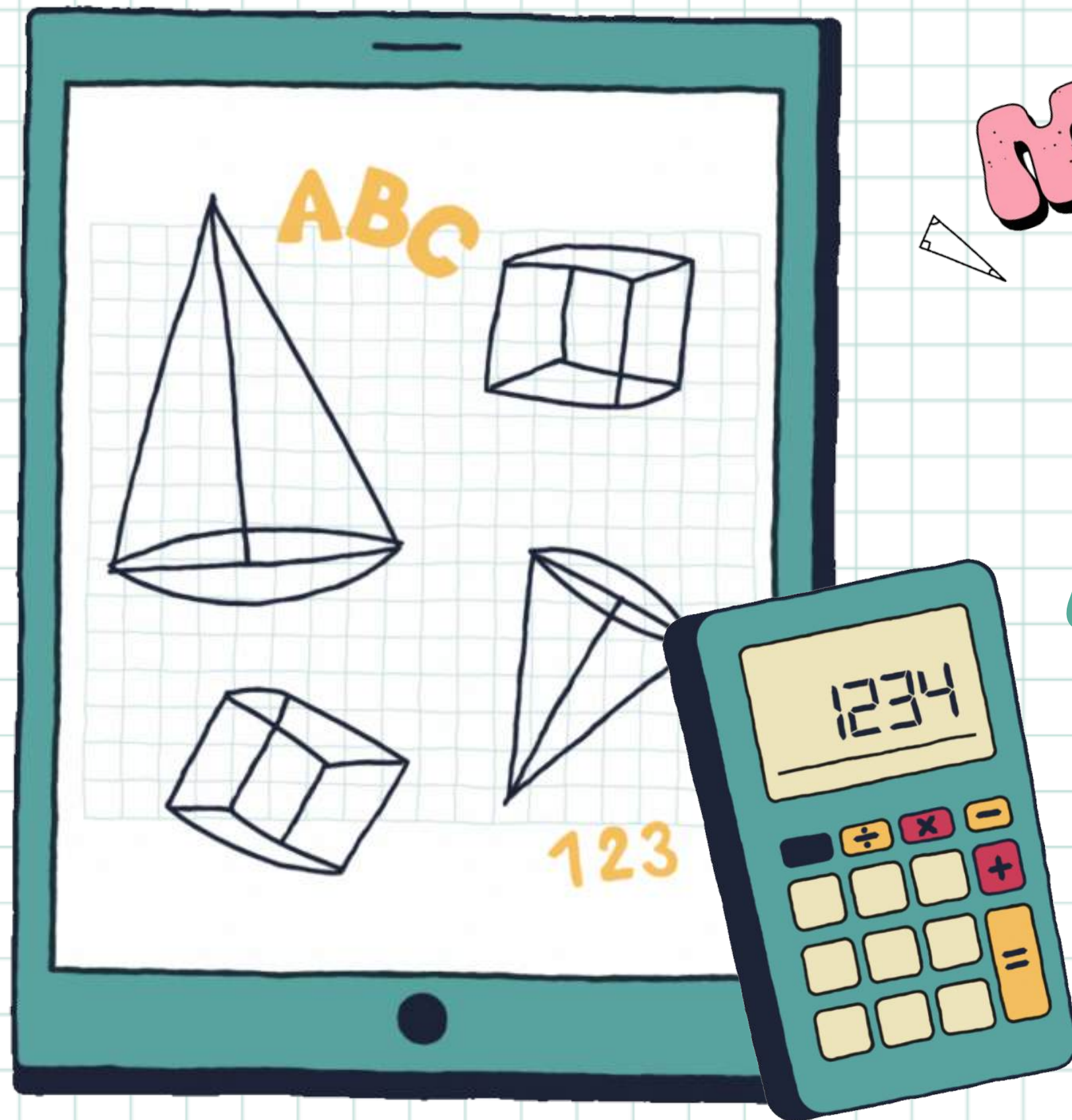


Exit Ticket Quiz

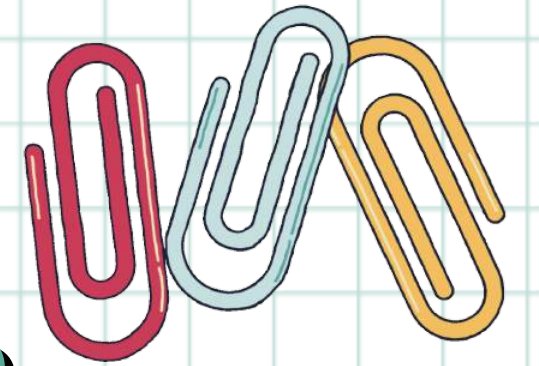
Q1: Count the squares — what is the area?
(See the shape with 8 squares)
Q2: A rectangle is 5 cm long and 3 cm wide.
Area = ? (Hint: $5 \times 3 = ?$)
Q3: True or False?
"Area and perimeter are the same thing."
Answers: Q1: 8 sq cm | Q2: 15 sq cm | Q3: FALSE



Source: AI-generated data. Replace or verify before use.



MATH $\left(\frac{a}{b}\right)^2$



Shabash!
You are an
AREA EXPERT!

- Well Done! You learned:
- ✓ What area means
 - ✓ How to count squares
 - ✓ The formula $L \times W$
 - ✓ Area vs Perimeter

